

Engineering Reboot 2026

Day 24 Workbook Supplement Expanded Exercise Guidance

Exercise 1 — Searching the Employee Directory

Objective

Learn to locate employee records using exact and partial searches while writing reusable functions.

Program: search_demo.py

Tasks

- Create `find_employee_by_id(employee_id)` that returns the matching employee dictionary or `None`.
- Create `find_employee_by_name(name)` that performs case-insensitive partial matching.
- Display a friendly message when no employee is found.

Engineering Discussion

IDs normally identify a single record, while names may return multiple results. Business systems rely on unique IDs for reliable lookups.

Exercise 2 — Filtering Employee Records

Objective

Create subsets of employee records without modifying the original dataset.

Program: filter_demo.py

Tasks

- Display Engineering employees.
- Display employees earning more than \$80,000.
- Display employees older than 35.
- Combine multiple filtering criteria.

Challenge

Create a reusable filtering function that accepts a predicate.

Engineering Discussion

Filtering creates a new collection based on business rules, similar to SQL WHERE clauses.

Exercise 3 — Sorting Employee Records

Objective

Present business information in useful orders using Python's sorting tools.

Program: sort_demo.py

Tasks

- Sort by name.
- Sort by salary (ascending and descending).
- Sort by age.
- Perform multi-level sorting by department then name.
- Display the top three highest-paid employees.

Engineering Discussion

The key function tells Python which value to compare when sorting complex objects such as dictionaries.

Mini Project — Employee Reporting System

Build a menu-driven reporting system supporting search by ID, search by name, Engineering employee report, high salary report, age report, department summary, payroll summary, highest-paid employee, lowest-paid employee, and average salary. Each menu option should call a dedicated reusable function.

Stretch Goals

- Total payroll using sum()
- Employee count per department
- Salary range searches
- Case-insensitive searching
- Top-N highest-paid employees
- Department average salary